

EX 4 – Demonstrate data visualization, histogram and multiple variable summaries

AIM :

To demonstrate data visualization, including histograms and multiple variable summaries, using a dataset in R/Python.

ALGORITHM :

1. Data Visualization

1. Import required libraries (Pandas, Matplotlib, Seaborn)
2. Load the dataset
3. Inspect the data using functions like head() and info()
4. Choose variables for visualization
5. Create basic plots (bar chart, line chart, scatter plot)

2. Histograms

1. Select a numerical variable
2. Plot a histogram to show data distribution
3. Adjust bins to improve clarity
4. Analyze shape (normal, skewed, uniform)

3. Multiple Variable Summaries

1. Use summary functions like:
 - describe() for statistical summary
 - groupby() for grouped analysis
2. Analyze relationships between variables
3. Use plots like:
 - Pair plot
 - Heatmap (for correlation)
4. Interpret trends and patterns

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/StudentsPerformance.csv")

print(df.head())
print(df.info())
print(df.describe())

print(df["math score"].describe())

plt.hist(df["math score"], bins=10)
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")
plt.show()

df[["math score","reading score","writing score"]].hist(bins=10, figsize=(10,5))
plt.show()

print(df.groupby("gender")[["math score","reading score","writing score"]].mean())

sns.boxplot(x="gender", y="math score", data=df)
plt.title("Math Score by Gender")
plt.show()
```

```
sns.heatmap(df[["math score","reading score","writing score"]].corr(), annot=True)
```

```
plt.title("Score Correlation")
```

```
plt.show()
```

OUTPUT :

```

gender race/ethnicity parental level of education lunch \
0 female group B bachelor's degree standard
1 female group C some college standard
2 female group B master's degree standard
3 male group A associate's degree free/reduced
4 male group C some college standard

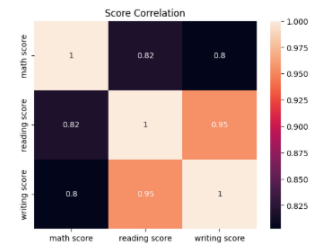
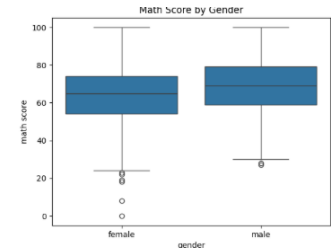
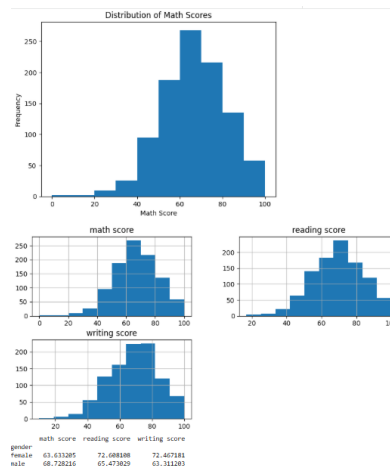
test preparation course math score reading score writing score
0 none 72 72 74
1 completed 69 90 88
2 none 90 95 93
3 none 47 57 44
4 none 76 78 75

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
# Column Non-Null Count Dtype
--
0 gender 1000 non-null object
1 race/ethnicity 1000 non-null object
2 parental level of education 1000 non-null object
3 lunch 1000 non-null object
4 test preparation course 1000 non-null object
5 math score 1000 non-null int64
6 reading score 1000 non-null int64
7 writing score 1000 non-null int64
dtypes: int64(3), object(5)
memory usage: 62.6+ KB
None

math score reading score writing score
count 1000.000000 1000.000000 1000.000000
mean 66.089000 69.169000 68.054000
std 15.16388 14.608192 15.195657
min 0.000000 17.000000 10.000000
25% 57.000000 59.000000 57.750000
50% 66.000000 70.000000 69.000000
75% 77.000000 79.000000 79.000000
max 100.000000 100.000000 100.000000

count 1000.000000
mean 66.089000
std 15.16388
min 0.000000
25% 57.000000
50% 66.000000
75% 77.000000
max 100.000000
Name: math score, dtype: float64

```



RESULT :

Thus the above program is executed and the result is obtained.